Course Title: **Artificial Intelligence in Education**

Course No. : ICT. Ed 482
Level: Bachelor
Program: BICTE
Semester: Eighth

Nature of course: Theoretical + Practical
Credit Hour: 2 (2 + 1)
Teaching Hour: 64(32+32)

1. Course Description

This course provides an introduction to Artificial Intelligence (AI) concepts and their applications in education. Students will explore key AI techniques such as search algorithms, knowledge representation, reasoning, machine learning, and natural language processing (NLP) while focusing on practical applications for educational systems.

2. General Objectives

The general objectives of this course are as follows:

- Understand fundamental AI concepts and their relevance to education.
- Explore AI techniques such as search, reasoning, and planning in the context of educational problem-solving.
- Apply machine learning and NLP methods in educational scenarios.
- Analyze the role of AI in adaptive learning and intelligent tutoring systems.
- Discuss ethical issues and emerging trends in AI for education.

3. Course Outlines:

Specific Objectives	Contents	Hours (Th+Pr)
• Understand the basics of AI and its applications in education. • Explore the historical development of AI. • Evaluate the effectiveness of AI applications in education. • Examine the use of learning agents in Education	**1. Introduction to AI** 1.1 Definition and evolution of AI 1.2 Applications of AI in education (adaptive learning, virtual tutors, etc.) 1.3 Importance of knowledge and learning agents in educational AI **Practical Works** • **Case Study Analysis**: Assign students to analyze real-world case studies of AI applications in education. They could explore platforms like Duolingo (adaptive learning) or IBM Watson Tutor and prepare a report on their AI-driven features and effectiveness.	3+3

• Implement and compare different search strategies • Understand the basic principles of uninformed search algorithms. • Understand the role of heuristics in search algorithms. • Understand the principles of adversarial search.	**2. Search Techniques** 2.1 Uninformed Search Techniques 2.1.1 Depth-first search, breadth-first search, depth-limited search, iterative deepening search 2.2 Heuristic Search Techniques 2.2.1 A* search, greedy best-first search, hill climbing 2.3 Adversarial Search 2.3.1 Minimax algorithm and alpha-beta pruning **<u>Practical Works</u>** • Implement a basic search algorithm to solve a real-world educational problem. For example: Implement DFS or BFS to search for books in the library catalog, helping to locate and retrieve books efficiently.	4+4
• Define propositional and predicate logic • Learn the components of logical expressions • Learn how rule-based systems represent knowledge as a set of rules and use inference mechanisms to make decisions or recommendations. • Understand statistical reasoning and its applications. • Develop automated grading systems	**3. Knowledge Representation and Reasoning 10** 3.1 Propositional and Predicate Logic 3.1.1 Syntax, semantics, and inference (forward chaining, backward chaining) 3.2 Rule-based Systems and Statistical Reasoning 3.2.1 Probability, Bayes' theorem, and belief networks 3.3 Applications in Education 3.3.1 Automated grading and lesson planning **<u>Practical Works</u>** • Implement a simple program to evaluate propositional logic expressions, including truth tables, to check the validity and satisfiability of logical statements. • Build a basic expert system for educational purposes, such as a career counseling system that suggests potential career paths based on student inputs (grades, interests, etc.). Use simple if-then rules to simulate expert advice. • Create a simple automated grading system that uses logical rules to grade multiple-choice or short-answer questions. Extend it to include probabilistic reasoning for handling uncertain or partial answers.	5+5
• Define machine learning and understand its significance in AI.	**4. Machine Learning and Neural Networks 12** 4.1 Overview of Machine Learning	6+6

• Compare and contrast different learning paradigms. • Understand the structure and components of neural networks. • Apply neural networks to educational problems.	4.1.1 Supervised, unsupervised, and reinforcement learning 4.2 Basics of Neural Networks 4.2.1 Architecture, training, and applications in education 4.3 Applications 4.3.1 Predicting student performance, learning analytics **Practical Works** • Develop a simple machine learning model using Python libraries like Scikit-learn. • Implement a simple supervised learning model (like linear regression or decision trees) to predict student grades based on features such as attendance, homework scores, and previous test scores. • Develop a machine learning model to predict student performance in upcoming exams based on historical performance data, demographics, and engagement metrics.	
• Explain the NLP. • Develop NLP applications to solve Educational problems.	**5. Natural Language Processing and Applications 8** 5.1 Overview of NLP 5.1.1 Parsing, text classification, and sentiment analysis 5.2 Applications in Education **5.2.1** Chat-bots, automated grading, and virtual assistants **Practical Works** • Create a basic chatbot for answering educational queries using Python's NLTK or Hugging Face.	4+4
• Discuss ethical implications and future trends in AI for education. • Understand the role of Generative AI in education. • Examine the ethical implications of emerging AI technologies.	**6. Ethical Considerations and Emerging Trends in AI** 6.1 Ethical considerations: Bias, Privacy, and Equity in AI systems 6.2 Emerging trends: Generative AI, Adaptive Learning Platforms, and Augmented Reality **Practical Works** • Use a generative AI model (like GPT or DALL·E) to create educational content, such as quizzes, lesson plans, or visual aids. Evaluate the	2+2

		quality and relevance of the generated content in a real educational context.	

4. Instructional Techniques

The instructional techniques for this course are divided into two groups. First group consists of general instructional techniques applicable to most of the units. The second group consists of specific instructional techniques applicable to specific units.

4.1 General Techniques

- Providing the reading materials to the students to familiarize the units.
- Lecture, question-answer, discussion, brainstorming, practical, hands-on labs, project based learning and flipped classrooms.

4.2 Specific Instructional Techniques

Unit	Activity and instructional techniques	Teaching Hours
1 to 6	• Organize coding labs where students can implement AI algorithms in languages like Python or tools like TensorFlow, PyTorch, or Scikit-learn. • Use Jupyter Notebooks for data analysis, visualizations, and model development, enabling students to work through real-world datasets. • Assign machine learning projects that require students to train AI models (e.g., predicting student performance, automating grading systems). • Include AI development tools for building neural networks or other machine learning models that can be directly applied to educational contexts (e.g., adaptive learning platforms). • Demonstrate various AI-based educational tools (e.g., virtual tutors, adaptive learning systems) and show how they work in a real classroom setting. • Encourage students to showcase their AI projects (e.g., predictive models, AI-based content creation) in a demo day format, allowing them to present their work to peers and instructors.	48

5. Evaluation (Internal Assessment and External Assessment):

Nature of course	Internal Assessment	External Practical Exam/Viva	Semester Examination	Total Marks
Theory	40%	20%	40%	100%

***Note**: Students must pass separately in internal assessment, external practical exam / viva and or semester examination.*

5.1 Evaluation for Part I (Theory): Internal Evaluation 40%

Internal evaluation will be conducted by course teacher based on following activities:

1) Attendance 5 points
2) Participation in learning activities 5 points
3) First assessment (written assignment) 10 points
4) Second assessment (Term examination) 10 points
5) Third assessment (Internal Practical Exam/Case Study) 10 points

Total 40 points

5.2 External Evaluation (Final Examination) 40%

Examination Division, office of the Dean, Faculty of Education will conduct final examination at the end of semester.

1) Objective type question (Multiple choice 10questionsx1mark) 10 marks
2) Short answer questions (6 questions x 5 marks) 30 marks

Total 40 marks

5.3 Evaluation for part II (practical) 20%

Nature of the course	Semester final examination by External Examiner	Total percent
Practical	100%	100

5.3.1 Practical Examination Evaluation Scheme

a) External assessment...100%
   i) Lab Report/Project Report..................20%
   ii) Laboratory work exam/Case 40%
   iii) VIVA..40%

6. Recommended books and reading materials (including relevant published articles in national and international journals)

Prescribed Text Book

- Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson

Reference Materials

- Holmes, W., Bialik, M., & Fadel, C. (2021). *Artificial intelligence for education: Promise and implications*. Center for Curriculum Redesign.
- Müller, V. C. (Ed.). (2020). *Ethics of artificial intelligence and robotics*. Stanford Encyclopedia of Philosophy.
- Holmes, W., Bialik, M., & Fadel, C. (2019). *AI in education: Challenges and opportunities*. Center for Curriculum Redesign.
- Shukla, N. (2019). *Deep learning for education*. Springer.
- Ng, A. (2018). *Machine learning yearning*. Deeplearning.ai.
- Goldman, R., Anderson, R. C., & Sullivan, P. S. (Eds.). (1999). *Artificial intelligence in education: Promises and implications for teaching and learning*. Springer.